Series 03 — Data wrangling and ANOVA

IEAP – RStudio

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#| label: setup#| include: falselibrary(tidyverse) ## Introduction

The present report analyzes pointing movements performed by 51 participants divided into three groups (ap, cp and pp). Each participant reached toward five target orientations (W, NW, N, NE and E), executing the movements either outward or inward, over two trials with up to three movement segments and up to three repetitions. Two dependent variables are analyzed: movement time and the number of velocity peaks, the latter reflecting how smooth the movement trajectory is. The data were loaded from the raw results file, converted to an analysis-ready format and aggregated to one mean value per subject and condition. Mixed analyses of variance were then fitted with GROUP as a between-subject factor and target orientation (ORI) and movement direction (DIR) as within-subject factors.

- **GitHub repository:** Isah1875/IEAP-Series03-RStudio
- **Course:** IEAP – RStudio, Series 03 (Instructor: Denis Mottet)
- **Author:** Isah Dadiga

1 Git workflow strategy

(to be completed later — how we used branches and commits)

2 Challenges and lessons learned

(to be completed later)

3 Data sources

(to be completed later — dataset description and URL)

4 Scientific references

(to be completed later — references with DOIs)

5 Questions

Data

The dataset was loaded from Results.txt (whitespace-separated, 9 variables).

#| label: load-dataresults <- read_table("data/Results.txt")

#| label: data-structureglimpse(results)

The dataset contains r nrow(results) observations and r ncol(results) variables, covering r n_distinct(results$SUBJ) subjects.

START Data wrangling

The experiment involved r n_distinct(results$SUBJ$) subjects divided into three groups : $ap(13 subjects), cp(19 subjects) and pp(19 subjects). Each subject performed pointing movements toward r n levels($ target orientations (E, N, NE, NW, W), in an outward or inward direction, over two trials (T01–T02), with up to three movement segments (ID1–ID3) recorded in up to three repetitions (R1–R3). The raw dataset contains r nrow(results) rows and r ncol(results) variables.

#| label: clean-dataresults_clean <- results %>% mutate(GROUP = as.factor(GROUP), TRIAL = as.factor(TRIAL), ORI = as.factor(ORI), DIR = as.factor(DIR), ID = as.factor(ID),

REP = as.factor(REP))tibble(check = c(“Missing values”, “Duplicated rows”), count = c(sum(is.na(results_clean)), sum(duplicated(results_clean))))

The dataset contains no missing values and no duplicated rows, so all r nrow(results_clean) observations are kept for analysis.

#| label: design-checkresults_clean %>% distinct(GROUP, SUBJ) %>% count(GROUP, name = “n_subjects”)results_clean %>% count(GROUP, name = “n_observations”)results_clean %>% count(SUBJ) %>% summarise(min = min(n), mean_n = mean(n), max = max(n))

The number of recorded movements per subject ranges from 80 to 180, so the design is unbalanced. This is expected in this type of experiment and does not prevent the ANOVA.

#| label: summary-statsresults_clean %>% group_by(ORI, DIR) %>% summarise(n = n(), mean_mt = mean(MovementTime), sd_mt = sd(MovementTime), mean_nvp = mean(NbVelPeaks), sd_nvp = sd(NbVelPeaks), .groups = “drop”)

Movement time and the number of velocity peaks vary across target orientations and movement directions. The ANOVA in the next section will test whether these differences are statistically significant.

Analysis of variance

Because each subject performed several movements in every condition, the raw rows are not independent observations. The data are therefore aggregated to one mean value per subject and condition, which yields r nrow(subj_means) subject-by-condition means before running the models.

#| label: aggregate-subjectssubj_means <- results_clean %>% group_by(GROUP, SUBJ, ORI, DIR) %>% summarise(mean_mt = mean(MovementTime), mean_nvp = mean(NbVelPeaks), .groups = “drop”)glimpse(subj_means)

A mixed ANOVA is fitted for movement time, with GROUP as between-subject factor and ORI and DIR as within-subject factors:

#| label: anova-mtmt_aov <- aov(mean_mt ~ GROUP * ORI * DIR + Error(SUBJ / (ORI * DIR)), data = subj_means)summary(mt_aov)

The same model is fitted for the number of velocity peaks:

#| label: anova-nvpnvp_aov <- aov(mean_nvp ~ GROUP * ORI * DIR + Error(SUBJ / (ORI * DIR)), data = subj_means)summary(nvp_aov)

Each Error: block of the output corresponds to one stratum of the design. The GROUP effect is tested in the Error: SUBJ block; ORI and its interaction with GROUP are tested in the Error: SUBJ:ORI block; DIR and its interaction with GROUP are tested in the Error: SUBJ:DIR block; and the remaining interactions are tested in the Error: SUBJ:ORI:DIR block. An effect is considered statistically significant when its p value, listed under Pr(>F), is below .05.

Results

Movement time. The ANOVA revealed a highly significant main effect of GROUP, $F(2, 48) = 53.79$, $p < .001$: the pp group was much slower ($M = 4.38$ s) than the ap group ($M = 1.51$ s) and the cp group ($M = 1.08$ s). Movement direction also had a significant main effect, $F(1, 48) = 11.55$, $p = .001$, with outward movements ($M = 2.60$ s) taking longer than inward movements ($M = 2.23$ s). This direction effect interacted with GROUP, $F(2, 48) = 4.09$, $p = .023$: the outward-minus-inward difference was much larger in the pp group (+0.77 s) than in the ap group (+0.17 s) and the cp group (+0.10 s). Target orientation did not significantly affect movement time overall, $F(4, 192) = 1.64$, $p = .166$, but interacted with GROUP, $F(8, 192) = 1.99$, $p = .049$, and entered a significant three-way GROUP × ORI × DIR interaction, $F(8, 192) = 2.22$, $p = .027$, indicating that the joint pattern of orientation and direction effects was not the same in all three groups.

Number of velocity peaks. The same model applied to the number of velocity peaks showed a highly significant GROUP effect, $F(2, 48) = 35.31$, $p < .001$: movements of the pp group contained far more velocity peaks ($M = 12.90$) than those of the ap group ($M = 3.96$) and the cp group ($M = 3.19$), reflecting much less smooth movements in the pp group. Target orientation had a significant main effect, $F(4, 192) = 12.13$, $p < .001$, with the highest number of peaks for the N orientation ($M = 8.22$) and the lowest for NW ($M = 6.12$), and this orientation effect interacted with GROUP, $F(8, 192) = 3.25$, $p = .002$. Movement direction had a smaller but significant effect, $F(1, 48) = 4.45$, $p = .040$, with more peaks for outward movements ($M = 7.50$) than inward movements ($M = 6.51$); the DIR × GROUP interaction was not significant, $F(2, 48) = 1.84$, $p = .169$, nor was the ORI × DIR interaction, $F(4, 192) = 2.17$, $p = .075$. The three-way GROUP × ORI × DIR interaction was significant, $F(8, 192) = 2.37$, $p = .019$.

Taken together, the pp group stands out on both dependent variables: slower, direction-sensitive movements and much less smooth trajectories, whereas the ap and cp groups behave similarly to each other. Orientation effects were absent for movement time but clearly present for the number of velocity peaks, suggesting that movement smoothness, more than movement duration, depends on target direction.

Figures

The figures below display the group means underlying the ANOVAs, with the target orientation on the horizontal axis (ordered from west to east) and one panel per movement direction.

#| label: cell-meanscell_means <- subj_means %>% group_by(GROUP, ORI, DIR) %>% summarise(mean_mt = mean(mean_mt), mean_nvp = mean(mean_nvp), .groups = “drop”) %>% mutate(ORI = fct_relevel(ORI, “W”, “NW”, “N”, “NE”, “E”))

#| label: fig-mt#| fig-cap: “Mean movement time by target orientation, movement direction and group.”ggplot(cell_means, aes(x = ORI, y = mean_mt, group = GROUP, color = GROUP)) + geom_line(linewidth = 1) + geom_point(size = 2) + facet_wrap(~ DIR) + labs(x = “Target orientation”, y = “Mean movement time (s)”, color = “Group”) + theme_minimal()

The movement time figure mirrors the ANOVA results: the pp group lies far above the two other groups in both panels, and the gap between outward and inward movements is visibly

larger for pp than for the ap and cp groups, which explains the significant GROUP × DIR interaction. The target orientation has little visible influence on movement time, consistent with its non-significant main effect.

#| label: fig-nvp#| fig-cap: “Mean number of velocity peaks by target orientation, movement direction and group.”ggplot(cell_means, aes(x = ORI, y = mean_nvp, group = GROUP, color = GROUP)) + geom_line(linewidth = 1) + geom_point(size = 2) + facet_wrap(~ DIR) + labs(x = “Target orientation”, y = “Mean number of velocity peaks”, color = “Group”) + theme_minimal()

The velocity peak figure shows the same clear group separation, together with a visible orientation effect: within each group, the number of peaks varies across target orientations and reaches its maximum around the N orientation. This matches the significant main effect of ORI and the significant GROUP × ORI interaction found for this variable.

6 Grading checklist

- ☐ README.md complete
- ☐ LICENSE file present
- ☐ Master Quarto document with sub-document includes
- ☐ PDF rendered and uploaded
- ☐ At least one commit per question
- ☐ Principled use of branches